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# 24CS206 25CSXXX – COMPUTER ORGANIZATION AND MICROPROCESSORS

**Hours Per Week :**

| L | T | P | SL | C |
|---|---|---|---|---|
| 3 | 2 | - | 3 | 4 |

PREREQUISITE KNOWLEDGE: Digital Logic Design.

**COURSE DESCRIPTION AND OBJECTIVES:**

This course covers the basics of modern Computer Organization and Microprocessor. The emphasis is on understanding the design of computer and its components. The student will learn the concepts of data representation, micro-operations, memory and input output organizations. It also deals with the basic architecture, various addressing modes, pin definitions, interfacing of 8086 with memory and peripheral devices.

## MODULE-1

18L+12T+18SL=48 Hours

### UNIT-1

Introduction, Register Transfer language, Register Transfer, Bus and Memory Transfers, Data Representation- Fixed Point Representation, Floating Point Representation.

Computer Arithmetic: Fixed-point arithmetic operations such as Addition and Subtraction, Multiplication Algorithms, and Division Algorithms.

### UNIT-2

Micro operations: Arithmetic, Logic, Shift Micro Operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design: Computer Register, Computer Instructions, Memory Reference Instructions. Register Reference Instructions, Input Output Reference Instructions, Interrupt and Instruction Cycle.

**PRACTICES:**

- Design a Common bus system for eight registers with eight bits each using multiplexers.
- Design a Common bus system for four registers with four bits each using three state gate buffers.
- A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.
- How many selection inputs are there in each multiplexer? What size of the multiplexers are needed? How many multiplexers are there in the bus?
- Perform arithmetic operations (+42) + (-13) and (-42) – (-13) in binary using signed 2's complement representation for negative numbers.
- Find the product using the Booth Multiplication Algorithm.

a. (9) x (13) b. (9) x (-13) c. (-9) x (13) d. (-9) x (-13)

- Perform the division of 27 and 4 using the Division algorithm.
- Design a 4-bit combinational circuit decrementer using 4 full adder circuits.
- Register A holds the 8-bit binary 11011001. Determine the B operand and the logic micro-operation to be performed to change the value in A to:

a) 01101101 b) 11111101

## MODULE-2

27L+18T+27SL=72 Hours

### UNIT 1

Memory Organization: Memory Hierarchy, Main Memory and its interfacing, Auxiliary Memory, Associative Memory, Cache Memory.

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer, Modes of Transfer, Priority Interrupts. Direct Memory Access (DMA).

**UNIT 2**

8086 microprocessor Architecture, Register model, Memory segmentation, Physical address generation, Addressing modes, Instruction set, Interrupts of 8086. Pin configuration of 8086.

**UNIT 3**

8086 Interfacing with static RAM and ROM chips, Address decoding techniques, 8255 PPI, Programmable Communication Interface (8251 USART), Programmable interrupt controller (8259).

**PRACTICES:**

- Programs on different Data Transfer Instructions using 8086.
- Arithmetic operations: Addition, Subtraction, Multiplication, Division using 8086.
- Programs to analyze different addressing modes of 8086.
- Program to sort the array of given numbers in ascending order.
- Interfacing of 8086 with RAM and ROM.
- Interfacing of 8086 with various IO devices using 8255 PPI.
- Interfacing of 8086 with 8251 USART.

**COURSE OUTCOMES:**

Upon successful completion of this course, students will have the ability to

| CO No. | Course Outcomes | Blooms Level | Module No. | Mapping with POs |
|---|---|---|---|---|
| 1 | Analyze the computer organization and system bus. | Analyze | 1 | 1,2,3, 9,10 |
| 2 | Design different digital circuits required to perform the micro operations. | Apply | 1 | 1,2,3, 9,10 |
| 3 | Design interface circuits for memory and peripheral, DMA, and communication devices. Compare various modes of transfer. | Analyze | 2 | 1,2,3,4,5, 6,7, 9,10 |
| 4 | Analyze the architecture of the 8086 microprocessor and its addressing modes. | Analyze | 2 | 1,2,3,4,5, 6,7, 9,10 |
| 5 | Analyze various interfaces for the 8086 microprocessor. | Analyze | 2 | 1,2,3,4,5, 6, 7, 9,10 |

| SDGs | IKS |
|---|---|
| SDG 7 (Affordable and Clean Energy) SDG 8 (Decent Work and Economic Growth) SDG 9 (Industry, Innovation, and Infrastructure) SDG 11(Sustainable Cities and Communities) | |

**TEXT BOOKS:**


- M. Morris Mano, “Computer System Architecture”, 3rd Edition update, Pearson, 2017.
- Douglas V.Hall, “Microprocessors and Interfacing”, 2nd edition, Tata McGraw Hill, 2017.


**REFERENCE BOOKS:**


- William Stallings, “Computer Organization & Architecture: Designing for Performance”, 11th Edition, Pearson, 2019.
- Advanced Microprocessors and Peripherals – A. K. Ray and K.M. Bhurchandani, TMH, 2nd Edition 2006.


**SKILLS:**

- ✓ *Analyze processor architectures, instruction set designs, and datapath execution.*
- ✓ *Evaluate cache hierarchy, memory management, and pipeline hazard resolution.*
- ✓ *Interface microprocessors with memory modules and high-speed I/O peripherals.*
- ✓ *Design and optimize high-throughput digital computing subsystems.*